

Arslan Khalid

Machine Learning Engineer

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EDUCATION

B.Sc. Electrical Engineering

University of Gujrat

CGPA: 3.12

Nov 2020 – Jul 2024

Relevant Courses:

- [Advanced ML Algorithms by Stanford University](#)
- [Supervised Machine Learning by Stanford University](#)
- [Django Web Framework by Meta](#)

SKILLS

Languages: Python · C++ · TypeScript · JavaScript · SQL

Libraries/Frameworks: PyTorch · TensorFlow · OpenCV · Hugging Face · LiveKit · FastAPI · Streamlit · React · Node.js

Machine Learning: LLMs · RAG Models · AI Agents · Prompt Engineering · RL · CV · NLP · Audio Synthesis (RVC)

Databases & Tools: PostgreSQL · Redis · Prisma · Docker · WebSockets · Git · GitHub

Soft Skills: Leadership · Team Coordination · Time Management · Critical Thinking · Problem Solving

EXPERIENCE

• Zylo Solutions (Remote)

AI Development Team Lead

Wyoming, USA

Nov 2025 – Present

- Lead a team of engineers in architecting and shipping production-grade agentic workflows, LLM applications, and real-time AI solutions.
- Streamlined project delivery and engineering standards, establishing robust CI/CD, testing, and evaluation pipelines for LLM prompts and models.

• Zylo Solutions (Remote)

Machine Learning Engineer

Wyoming, USA

May 2025 – Nov 2025

- Developed and optimized distributed ML/AI pipelines, integrating FastAPI backends with state-of-the-art NLP and computer vision models.
- Enhanced API responsiveness and reduced model inference latency by implementing efficient caching and model compilation techniques.

• GulzarSoft (On-site)

Machine Learning Engineer

Gujrat, Pakistan

Nov 2024 – Apr 2025

- Designed and implemented a collaborative filtering model for a car recommendation system, enhancing recommendation accuracy by 30%.

• Coders Launchpad (Remote)

Machine Learning Engineer

Dubai, UAE

Dec 2023 – Apr 2024

- Developed a voice-to-voice conversion model for the Quran Project using RVC technology to transform users' voices while maintaining high-quality audio synthesis.
- Conducted rigorous training and optimization of the voice modulation models to ensure high accuracy, pronunciation clarity, and natural-sounding synthesis.

• Intelligent Systems Lab (On-site)

Machine Learning Intern

Gujrat, Pakistan

Sep 2023 – Oct 2023

- Developed a machine learning model for depth estimation and path planning for an autonomous quadruped robot.
- Integrated stereo vision cameras and various onboard sensors to enable obstacle detection and real-time navigation.

PROJECTS

- **Stereo Vision Navigation with RL:** Developed an autonomous quadruped robot for path planning and navigation using reinforcement learning. Integrated obstacle detection and avoidance with advanced machine learning algorithms.
- **Jarvis — Real-Time AI Voice Assistant**
LiveKit, Python, WebSockets, Wake-Word Detection
 - Architected a persistent, low-latency AI voice assistant leveraging LiveKit real-time infrastructure.
 - Trained and deployed custom wake-word detection models, optimizing streaming connection latency to achieve near-instantaneous conversational responsiveness.
 - Bridged continuous audio streaming with backend AI processing pipelines, enabling seamless and natural user interactions.
- **Cortex — LLM Reasoning Visualization Tool**
React, TypeScript, Python, FastAPI, WebSockets
 - Designed and deployed a full-stack software system dedicated to tracking, logging, and visually mapping the internal reasoning chains of Large Language Models (LLMs).
 - Built a highly responsive frontend dashboard and robust backend architecture to process and display complex AI inference steps in real-time.
 - Enabled developers to effectively debug, audit, and optimize LLM prompts and agent behaviors through comprehensive visual analytics.
- **AI-FSL — EdTech Curriculum Generator**
Python, OpenAI API, Structured Outputs, JSON Schema
 - Created an automated lesson plan generator and comprehensive French immersion platform adopted by Canadian school boards.
 - Engineered a specialized generation engine utilizing strict structured parameters and rule-based logic to perfectly align AI outputs with official educational standards.
 - Streamlined the creation of highly specialized educational materials, significantly reducing curriculum preparation time for educators.